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Part-A : Conceptual foundation

(1) What are Ensemble learning Methods and why are they effective?

→ Ensemble learning is a machine learning technique that combins multiple models to produce a better prediction than a single model. It improves accuracy reduce errors, minimizes overfiting and provides more reliable and robust predictions.

(2) Explain the difference between Bagging, Boosting and Voting.

→ Bagging trains multiple models independently on different random samples and combins their predictions to reduce variance.

Boosting trains models sequentially, where each new model focuses on correcting the errors of the previous one.

Voting combines predictions from multiple trained models using majority voting (hard voting) or average probabilities (soft voting).

(3) Explain voting classifier and stacking Ensemble.

→ A voting classifier combines predictions from multiple classifier using majority voting (hard voting) or probability averaging (soft voting).

A stacking Ensemble combines predictions from several base models and uses a meta-model to make the final prediction.

(4) What is the Bias-Variance Trade-off and how do ensemble methods address it?

→ The Bias-Variance Trade-off is the balance between underfiting (high bias) and overfitting (high variance). Bagging mainly reduces variance, Boosting reduces bias and combining multiple models helps achieve better generalization.

(5) Compare AdaBoost, Gradient Boosting, LightGBM and XGBoost conceptually.

→ AdaBoost : Focuses on misclassified samples by assigning them higher weights in subsequent.

Gradient Boosting : Minimizes prediction errors by fitting each new model to the residual errors.

LightGBM : A fast memory-efficient gradient boosting algorithm that uses histogram based learning and leaf-wise tree growth.

XGBoost : An optimized gradient boosting algorithm with regularization, parallel processing and excellent accuracy and robustness.